Software Requirements Specification

Software Requirements Specification for Learning Management System (LMS)

Version 1.0

Prepared by

Inshal Nasir 2023-BSE-038 @23-22412-038@se.fjwu.edu.pk

Instructor: Engr. Sumaira Shaukat

Course: Software Engineering Fundamentals

Lab Section: Section A

Date: June 15, 2024

CONTENTS ..... ERROR! BOOKMARK NOT DEFINED.

REVISIONS 2

1 INTRODUCTION ..... ERROR! BOOKMARK NOT DEFINED.

1.1 Document Purpose ..... **Error! Bookmark not defined.**

1.2 Product Scope ..... **Error! Bookmark not defined.**

1.3 Intended Audience and Document Overview ..... **Error! Bookmark not defined.**
1.4 Definitions, Acronyms and Abbreviations ..... **Error! Bookmark not defined.**
1.5 Document Conventions ..... **Error! Bookmark not defined.**
1.6 References and Acknowledgments ..... **Error! Bookmark not defined.**
2 OVERALL DESCRIPTION ..... ERROR! BOOKMARK NOT DEFINED.
2.1 Product Overview ..... **Error! Bookmark not defined.**
2.2 Product Functionality ..... **Error! Bookmark not defined.**
2.3 Design and Implementation Constraints ..... **Error! Bookmark not defined.**
2.4 Assumptions and Dependencies ..... **Error! Bookmark not defined.**
3 SPECIFIC REQUIREMENTS ..... ERROR! BOOKMARK NOT DEFINED.
3.1 External Interface Requirements ..... **Error! Bookmark not defined.**
3.2 Functional Requirements ..... **Error! Bookmark not defined.**
3.3 Use Case Model ..... **Error! Bookmark not defined.**
4 OTHER NON-FUNCTIONAL REQUIREMENTS ..... ERROR! BOOKMARK NOT DEFINED.
4.1 Performance Requirements ..... **Error! Bookmark not defined.**
4.2 Safety and Security Requirements ..... **Error! Bookmark not defined.**
4.3 Software Quality Attributes ..... **Error! Bookmark not defined.**
5 OTHER REQUIREMENTS ..... ERROR! BOOKMARK NOT DEFINED.
APPENDIX A – DATA DICTIONARY ..... ERROR! BOOKMARK NOT DEFINED.
APPENDIX B - GROUP LOG ..... ERROR! BOOKMARK NOT DEFINED.

Revisions

Version	Primary authors	Description of version	Date Completed
0.1	Alishba Zainab	Initial version	1/6/2024
0.2	Inshal Nasir	Included use case models	5/6/2024
0.3	Lujain Zia	Revised interface specifications	10/6/2024
0.4	Fatima Kashif	Updated non-functional requirements	12/6/2024
1.1	Fatima Kashif	Initial release	15/6/2024

Introduction

1.1 Document Purpose

This document specifies the software requirements for the development of a Learning Management System (LMS) for a university. The LMS will streamline the process of delivering

educational courses, managing student records, and facilitating communication between students and instructors.

1.2 Product Scope

The LMS aims to enhance the educational experience by providing an integrated platform for course management, assignment submission, grading, and student-instructor communication. Key benefits include:

- **Increased Efficiency in Administrative Tasks**: Streamlining processes such as course creation, student enrollment, and assignment management.
- **Improved Accessibility to Course Materials**: Allowing students to access learning resources anytime and anywhere via a web-based platform.
- **Enhanced Interaction Among Users**: Facilitating communication through forums, messaging, and announcements to promote a collaborative learning environment.
- **Comprehensive Reporting and Analytics**: Providing insights into student performance and course effectiveness to help educators improve their teaching strategies.
- **Integration with Third-Party Tools**: Incorporating tools like plagiarism detection software to maintain academic integrity.
- **Scalability and Flexibility**: Supporting the university's growth and evolving needs by being adaptable to new requirements and user demands.
- **Security and Compliance**: Ensuring data protection and compliance with relevant regulations such as FERPA.

This comprehensive platform aims to provide a seamless and efficient educational experience for both students and instructors.

1.3 Intended Audience and Document Overview

This document is intended for developers, project managers, marketing staff, users, testers, and documentation writers involved in the LMS project. It provides a comprehensive overview of the system requirements, functionality, and constraints. The document is organized to facilitate a sequential understanding, starting from high-level descriptions to detailed specifications.

1.4 Definitions, Acronyms, and Abbreviations

- LMS: Learning Management System
- SRS: Software Requirements Specification
- UI: User Interface
- API: Application Programming Interface
- IEEE: Institute of Electrical and Electronics Engineers.

1.5 Document Conventions

This document follows the IEEE formatting requirements. Text is presented in Arial font size 11, single-spaced, with 1” margins. Section and subsection titles are formatted as per the template.

1.6 References and Acknowledgments

- IEEE Standard for Software Requirements Specifications
- University Software Engineering Course Materials

Overall Description

2.1 Product Overview

The LMS is designed to integrate various educational tools into a cohesive platform. It will replace outdated systems and provide new functionalities for course management and student engagement. The system includes components for course creation, student enrollment, assignment tracking, and communication tools.

2.2 Product Functionality

- **User Management:** Registration, login, and role-based access control
- **Course Management:** Creation, modification, and deletion of courses
- **Content Delivery:** Uploading and organizing course materials
- **Assignment Management:** Submission, grading, and feedback
- **Communication Tools:** Forums, messaging, and announcements
- **Reporting and Analytics:** Tracking student performance and course statistics

2.3 Design and Implementation Constraints

- **Technological Constraints:** The system must be built using the COMET method and UML modeling language.
- **Hardware Constraints:** The system should operate on standard university hardware and integrate with existing IT infrastructure.
- **Security Constraints:** Must comply with university data protection policies.

2.4 Assumptions and Dependencies

- The LMS will rely on the university's existing network infrastructure.
- Assumes availability of third-party tools for certain functionalities (e.g., plagiarism detection).

Specific Requirements

3.1 External Interface Requirements

User Interfaces

The LMS will feature a web-based interface accessible via standard web browsers. It will include dashboards for students and instructors, each tailored to their specific needs.

Hardware Interfaces

- Servers: The system will run on university servers.
- User Devices: Compatible with desktops, laptops, tablets, and smartphones.

Software Interfaces

- Integration with university databases for student information
- API for third-party tool integration
- Single Sign-On (SSO)
- Email and Notification Services
- Learning Tools Interoperability (LTI)
- Virtual Classroom Integration

3.2 Functional Requirements

- **User Management**
  - **F1**: The system shall allow users to register and log in.
  - **F2**: The system shall provide role-based access control.
  - **F3**: The system shall allow users to reset their passwords through a secure process.
  - **F4**: The system shall allow administrators to manage user roles and permissions.
- **Course Management**
  - **F5**: The system shall allow instructors to create, modify, and delete courses.
  - **F6**: The system shall allow instructors to upload and organize course materials, including documents, videos, and links.
  - **F7**: The system shall allow students to view and enroll in available courses.
  - **F8**: The system shall provide a search function to find courses based on keywords, categories, or instructors.
- **Assignment Management**
  - **F9**: The system shall allow students to submit assignments online.
  - **F10**: The system shall allow instructors to grade assignments and provide feedback.
  - **F11**: The system shall notify students of assignment deadlines and submission confirmations.

- **F12**: The system shall allow for the downloading and uploading of bulk assignments by instructors.

- **Communication Tools**

  - **F13**: The system shall provide forums for course-related discussions.
  - **F14**: The system shall allow for private messaging between users.
  - **F15**: The system shall allow instructors to post announcements that are visible to all students enrolled in a course.
  - **F16**: The system shall send email notifications for new messages, announcements, and forum posts.

- **Reporting and Analytics**

  - **F17**: The system shall provide reports on student performance, including grades and participation.
  - **F18**: The system shall provide analytics on course effectiveness and engagement.
  - **F19**: The system shall allow instructors to export reports in various formats (e.g., PDF, Excel).
  - **F20**: The system shall provide real-time dashboards for monitoring system usage and performance.

- **Content Delivery**

  - **F21**: The system shall allow instructors to schedule the release of course materials.
  - **F22**: The system shall support multimedia content, including video, audio, and interactive elements.
  - **F23**: The system shall allow students to download course materials for offline access.
  - **F24**: The system shall provide a content preview feature for materials before they are published.

- **Assessment and Quizzes**

  - **F25**: The system shall allow instructors to create and manage quizzes and exams.
  - **F26**: The system shall allow students to take quizzes and exams online.
  - **F27**: The system shall provide automatic grading for multiple-choice and true/false questions.
  - **F28**: The system shall allow for manual grading of essay-type questions.

- **Accessibility and Usability**

  - **F29**: The system shall comply with accessibility standards to support users with disabilities.
  - **F30**: The system shall provide customizable themes and layouts to enhance usability.
  - **F31**: The system shall offer multilingual support for international users.

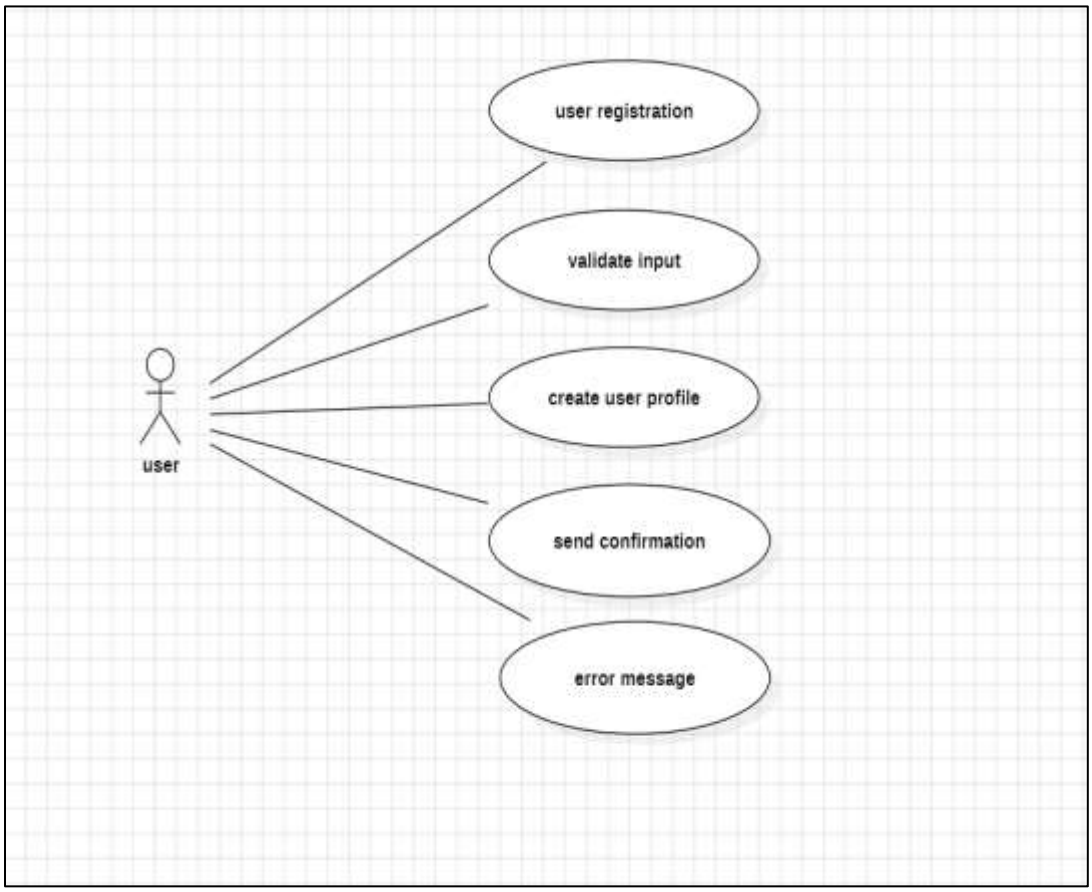
- **System Administration**

  - **F32**: The system shall provide tools for administrators to monitor and manage system performance.
  - **F33**: The system shall allow for regular data backups and recovery options.
  - **F34**: The system shall provide audit logs for tracking user activities and system changes.
  - **F35**: The system shall allow administrators to configure system settings and preferences.

3.3 Use Case Model

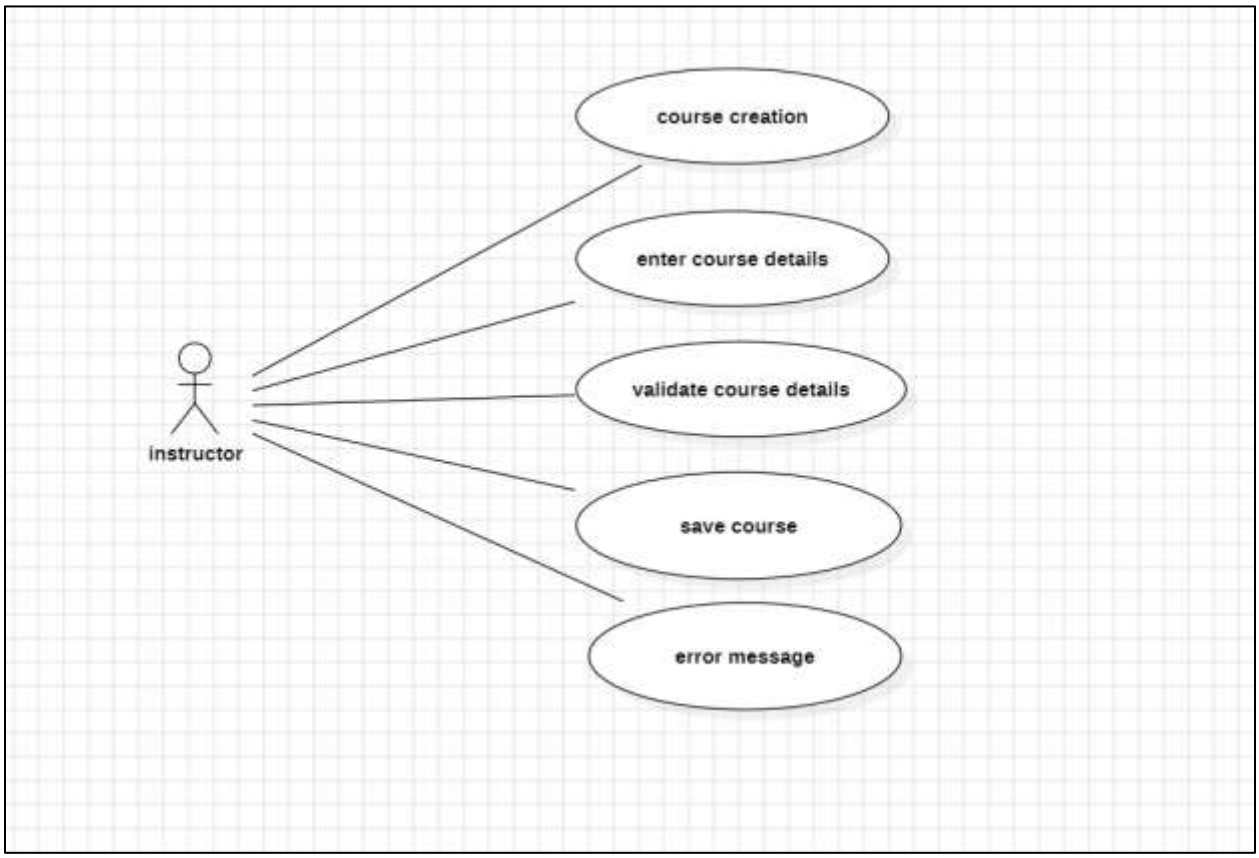
Use Case #1: User Registration

- **Author:** Inshal Nasir
- **Purpose:** To allow new users to register.
- **Requirements Traceability:** F1
- **Priority:** High
- **Preconditions:** User must have a valid email address.
- **Post conditions:** User account is created and ready for use.
- **Actors:** New user
- **Flow of Events:**
  - **Basic Flow:** User fills in registration form, submits, receives confirmation email.
  - **Alternative Flow:** If email is already in use, display error message.



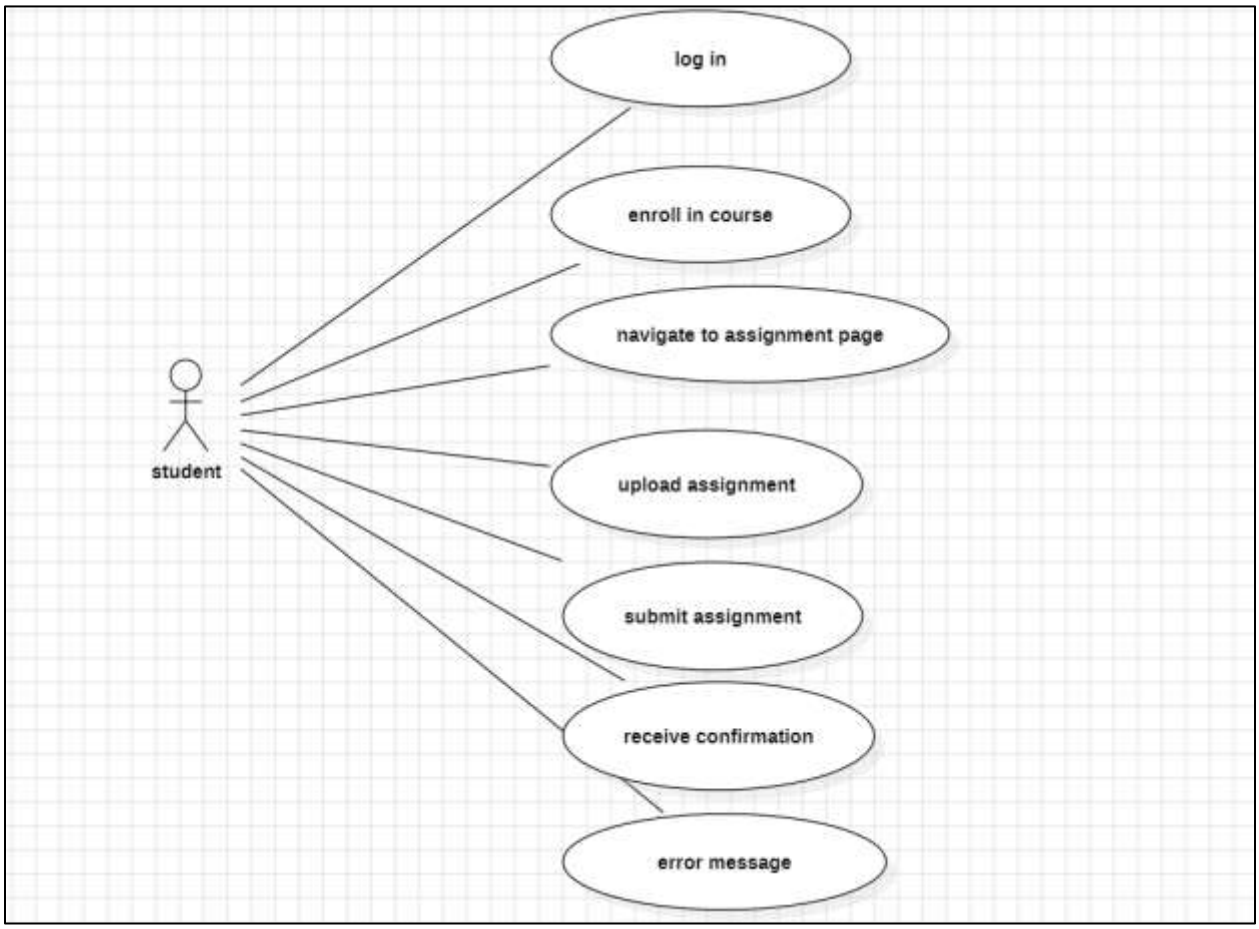
Use Case #2: Course Creation

- **Author:** Alishba Zainab
- **Purpose:** To enable instructors to create new courses.
- **Requirements Traceability:** F3
- **Priority:** High
- **Preconditions:** Instructor must be logged in.
- **Post conditions:** Course is created and listed.
- **Actors:** Instructor
- **Flow of Events:**
    - **Basic Flow:** Instructor enters course details and submits.
    - **Alternative Flow:** If mandatory fields are empty, display error message.



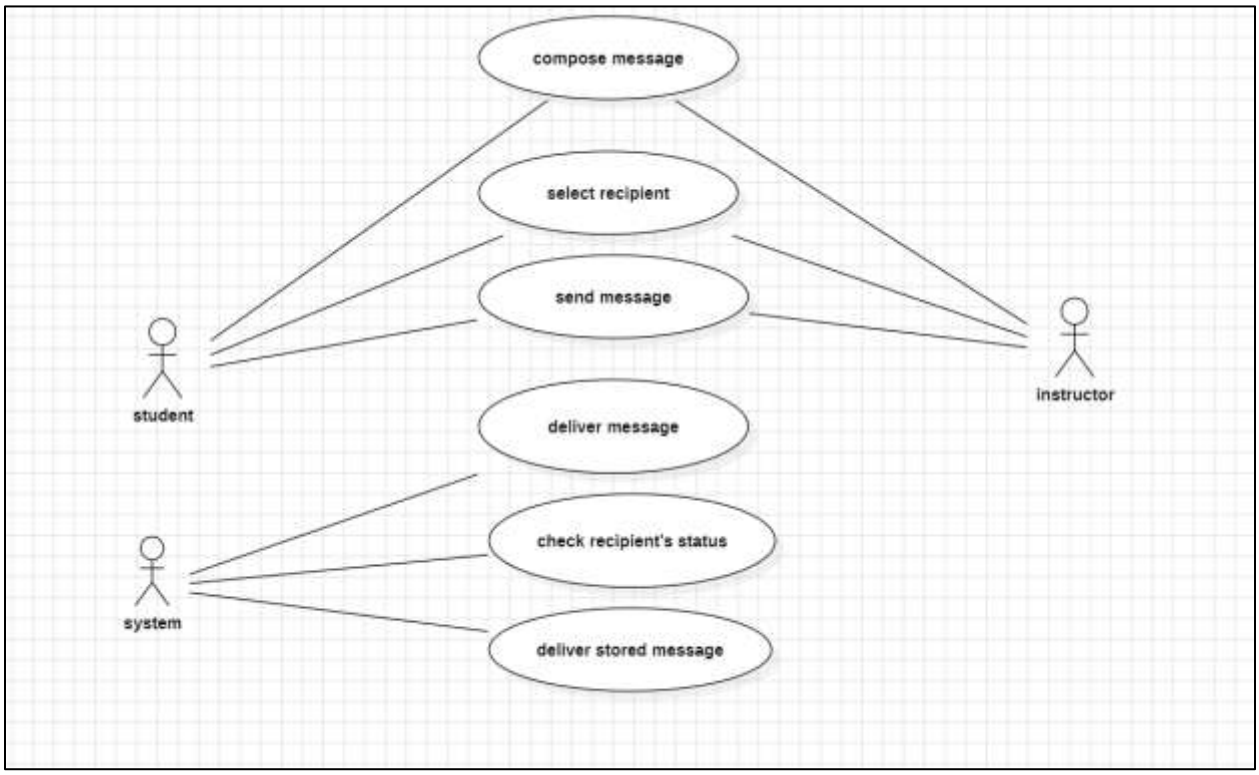
Use Case #3: Assignment Submission

- **Author:** Fatima Kashif
- **Purpose:** To allow students to submit their assignments.
- **Requirements Traceability:** F9
- **Priority:** High
- **Preconditions:** Student must be logged in and enrolled in the course.
- **Postconditions:** Assignment is submitted, and confirmation is provided to the student.
- **Actors:** Student
- **Flow of Events:**
    - **Basic Flow:** Student navigates to the assignment page, uploads the assignment file, and submits it.
    - **Alternative Flow:** If the file format is incorrect, an error message is displayed.



Use Case #4: Messaging

- **Author:** Inshal Nasir
- **Purpose:** To enable communication between users.
- **Requirements Traceability:** F14
- **Priority:** Medium
- **Preconditions:** Users must be logged in.
- **Postconditions:** Message is sent and received by the intended recipient.
- **Actors:** Student, Instructor
- **Flow of Events:**
    - **Basic Flow:** User composes a message, selects the recipient, and sends the message.
    - **Alternative Flow:** If the recipient is offline, the message is stored and delivered when they log in.



Use Case #5: Reporting and Analytics

- **Author:** Lujain Zia
- **Purpose:** To track student performance and generate reports.
- **Requirements Traceability:** F17
- **Priority:** Medium
- **Preconditions:** User must have appropriate access rights.
- **Postconditions:** Report is generated and available for review.
- **Actors:** Admin, Instructor
- **Flow of Events:**
    - **Basic Flow:** User selects report parameters, generates the report, and views the results.
    - **Alternative Flow:** If data is incomplete, display error message.



Other Non-functional Requirements

4.1 Performance Requirements

- **P1:** The system shall load the dashboard within 3 seconds.
- **P2:** The system shall handle up to 1000 concurrent users without performance degradation.

4.2 Safety and Security Requirements

- **S1:** The system shall enforce HTTPS for all communications.
- **S2:** User data shall be encrypted in storage and transit.

4.3 Software Quality Attributes

- **Reliability:** The system shall have an uptime of 99.9%.
- **Maintainability:** The system shall be modular to allow easy updates and maintenance.

Other Requirements

- **Internationalization:** The system shall support multiple languages.
- **Legal Requirements:** Must comply with FERPA regulations.

Appendix A – Data Dictionary

Item	Description	Type	Possible Values	Operations	Requirements
userID	Unique identifier for users	String	Alphanumeric	Create, Read, Update, Delete	F1: Registration, F2: Login
courseID	Unique identifier for courses	String	Alphanumeric	Create, Read, Update, Delete	F3: Course Management
assignmentID	Unique identifier for assignments	String	Alphanumeric	Create, Read, Update, Delete	F9: Assignment Management
username	Name of the user	String	Alphanumeric	Read, Update	F1: Registration, F2: Login
password	User's password	String	Alphanumeric (encrypted)	Create, Update	F1: Registration, F2: Login
role	Role of the user in the system	Enum	Admin, Instructor, Student	Read, Update	F2: Role-based Access Contro
courseName	Role of the user in the system	String	Alphanumeric	Create, Read, Update	F5: Course Management
courseDesc	Name of the course	String	Alphanumeric	Create, Read, Update	F5: Course Management
assignmentTitle	Title of the assignment	String	Alphanumeric	Create, Read, Update	F9: Assignment Management
assignmentDesc	Description of the assignment	String	Alphanumeric	Create, Read, Update	F9: Assignment Management

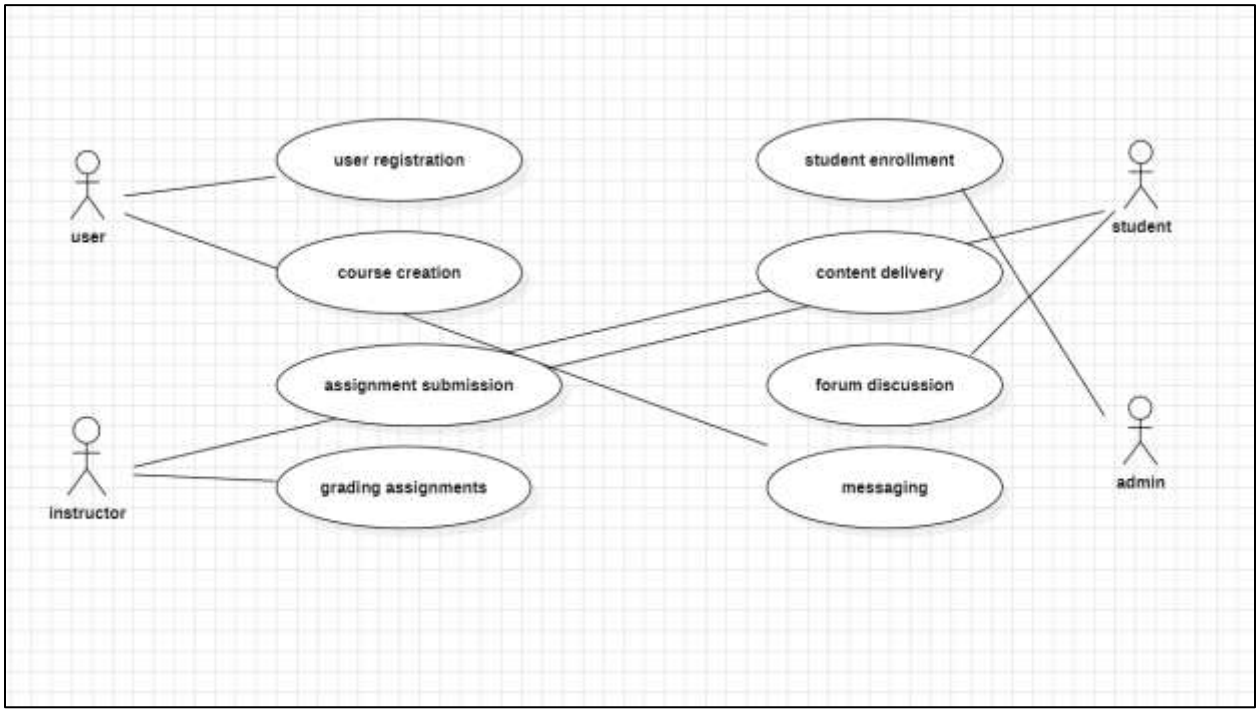
submissionDate	Date of assignment submission	Date	YYYY-MM-DD	Create, Read, Update	F9: Assignment Management
Grade	Grade for the assignment	Float	0.0 - 100.0	Create, Read, Update	F9: Assignment Management
Feedback	Feedback on assignment	String	Alphanumeric	Create, Read, Update	F9: Assignment Management
messageID	Unique identifier for messages	String	Alphanumeric	Create, Read, Update, Delete	F14:Messaging
messageContent	Content of the message	String	Alphanumeric	Create, Read, Update	F14:Messaging
forumID	Unique identifier for forum posts	String	Alphanumeric	Create, Read, Update, Delete	F13:Forum Discussion
forumContent	Content of the forum post	String	Alphanumeric	Create, Read, Update	F13:Forum Discussion
reportID	Unique identifier for reports	String	Alphanumeric	Create, Read, Update, Delete	F17:Reporting and Analytics
reportData	Data included in the report	String	Alphanumeric	Create, Read, Update	F17:Reporting and Analytics
enrollmentID	Unique identifier for enrollments	String	Alphanumeric	Create, Read, Update, Delete	F7:Student Enrollment
studentID	Unique identifier for students	String	Alphanumeric	Read, Update	F2: Role-based Access Control
instructorID	Unique identifier for instructors	String	Alphanumeric	Read, Update	F2: Role-based Access Control

Appendix B - Group Log

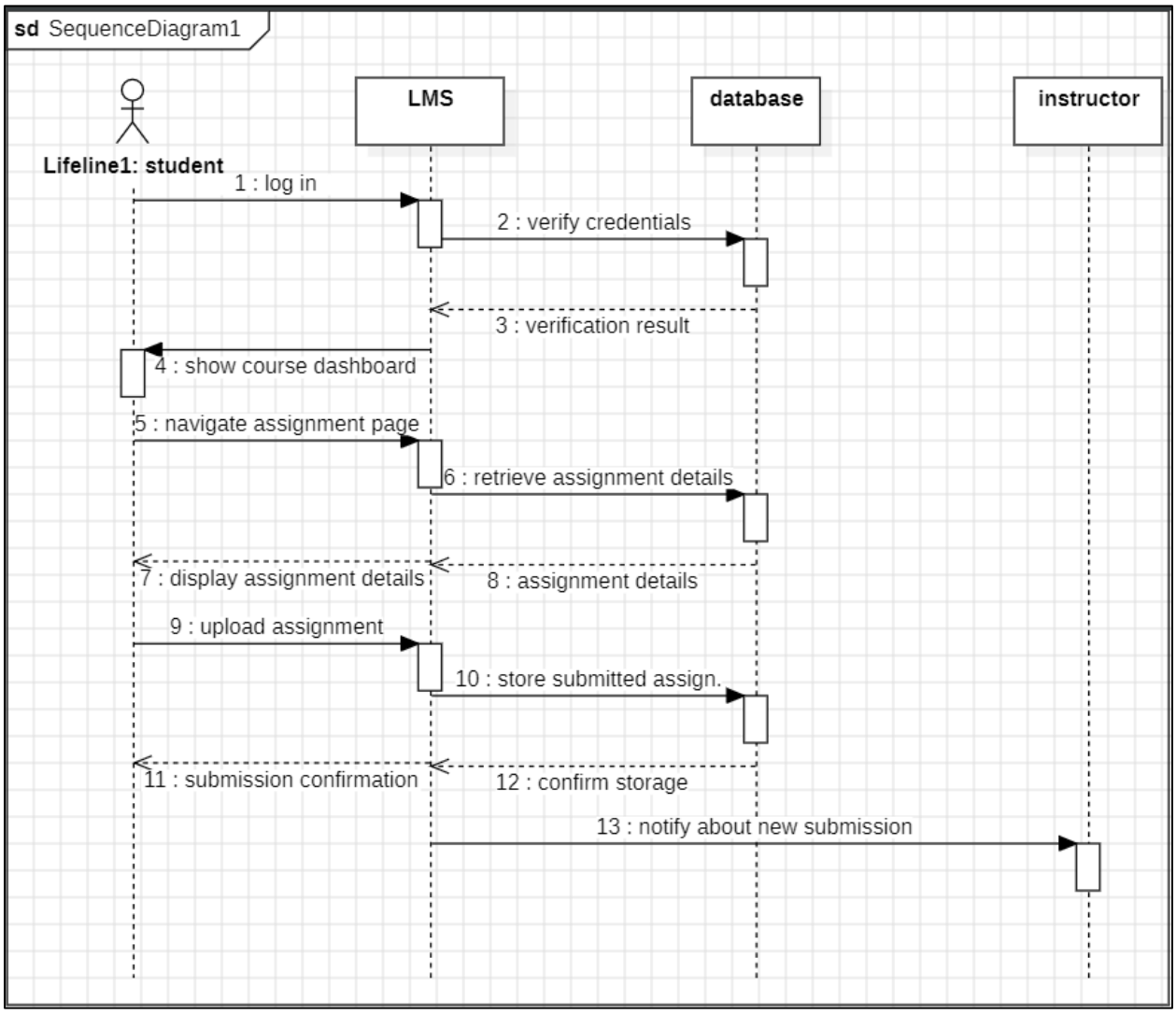
- **Meeting 1:** Discussed project scope and initial requirements.
- **Meeting 2:** Assigned tasks and started drafting SRS.
- **Meeting 3:** Reviewed and finalized initial draft of SRS.

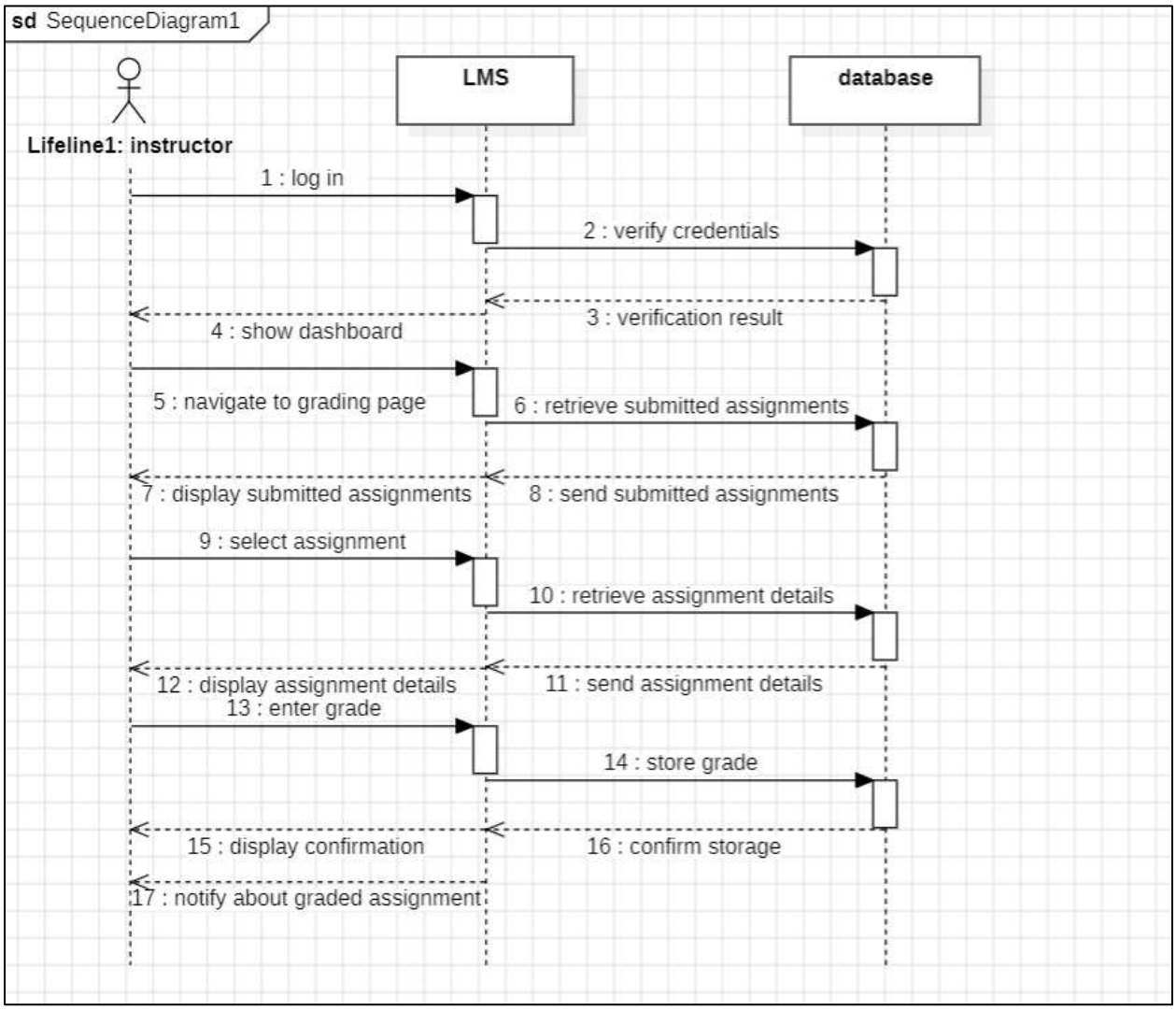
Diagrams:

Use case:

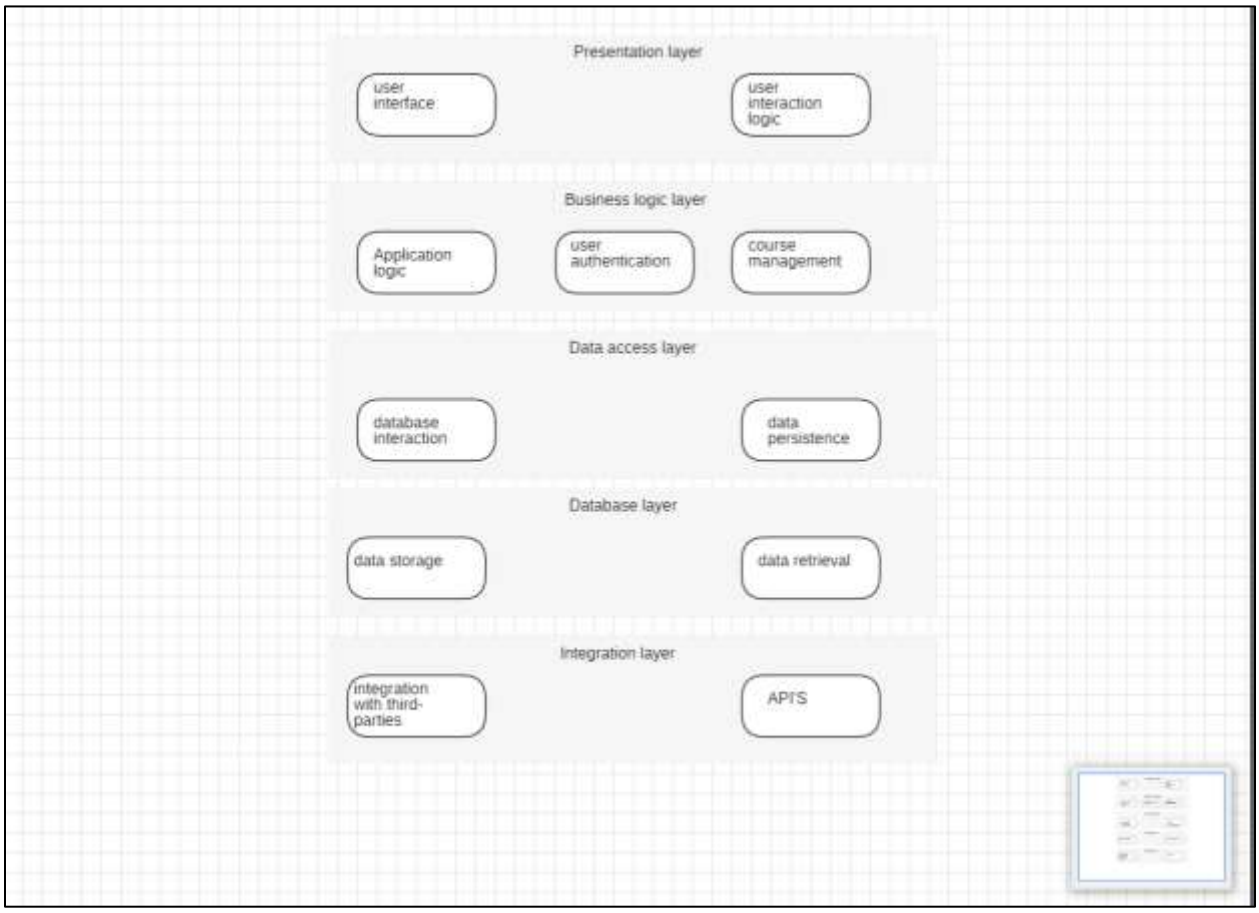


Sequence diagrams:

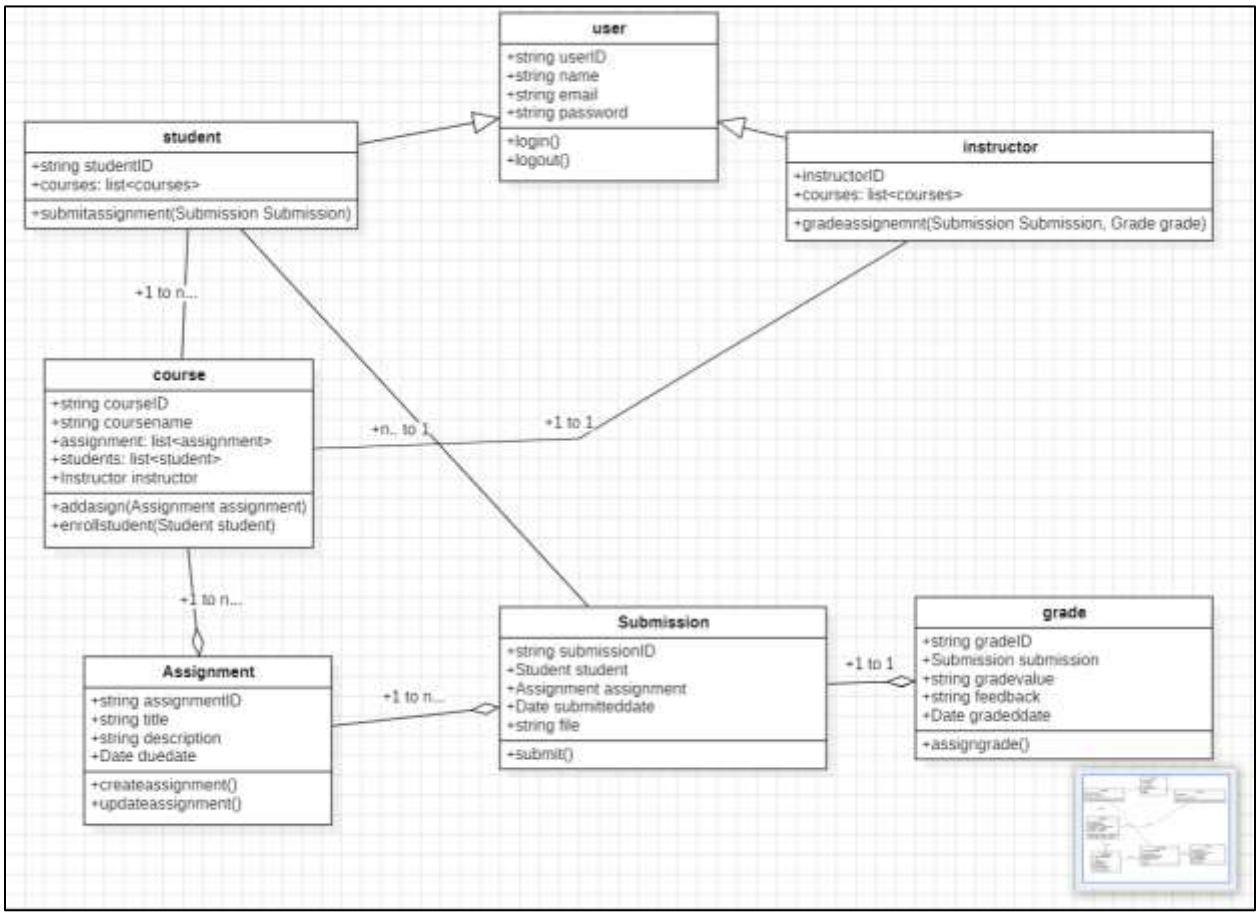

sd SequenceDiagram1
LMS
database
instructor
Lifeline1: student
1 : log in
2 : verify credentials
3 : verification result
4 : show course dashboard
5 : navigate assignment page
6 : retrieve assignment details
7 : display assignment details
8 : assignment details
9 : upload assignment
10 : store submitted assign.
11 : submission confirmation
12 : confirm storage
13 : notify about new submission



Architecture diagram:



Class diagram:


user
+string userID
+string name
+string email
+string password
+login()
+logout()
student
+string studentID
+courses: list<courses>
+submitassignment(Submission Submission)
instructor
+instructorID
+courses: list<courses>
+gradeassignemnt(Submission Submission, Grade grade)
+1 to n...
course
+string courseID
+string coursename
+assignment: list<assignment>
+students: list<student>
+Instructor instructor
+addasign(Assignment assignment)
+enrollstudent(Student student)
+n.. to 1
+1 to 1
+1 to n...
Assignment
+string assignmentID
+string title
+string description
+Date duedate
+createassignment()
+updateassignment()
+1 to n...
Submission
+string submissionID
+Student student
+Assignment assignment
+Date submitteddate
+string file
+submit()
+1 to 1
grade
+string gradeID
+Submission submission
+string gradevalue
+string feedback
+Date gradeddate
+assigngrade()